第五章 CPU 调度

1. Why is it important for the scheduler to distinguish I/O-bound programs from CPU-bound programs?

I/O-bound programs spend most time waiting for I/O opts, and they need short and frequent bursts for CPU.

CPU-bound programs use CPU in most time and they need longer CPU bursts.

The scheduler can improve the utilization of CPU, cut off response time, and increase the throughput by distinguishing the two kinds of programs.

2. Discuss how the following pairs of scheduling criteria conflict in certain settings.

a. CPU utilization and response time

b. Average turnaround time and maximum waiting time

c. I/O device utilization and CPU utilization

a. Maximizing CPU utilization means use longer time shices or scheduling CPU-bound process effucienthy, but it may delay short burst processes and increasing response time.

b. In SJF scheduling, the strategy of short jobs first can reduce average turnaround time, but long jobs may starve, increasing their maximum waiting time.

c. If sets high priority to I/O devie, CPU utilization will decrease; and vice versa.

3. Consider the following set of processes, with the length of the CPU-burst time given in milliseconds:

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

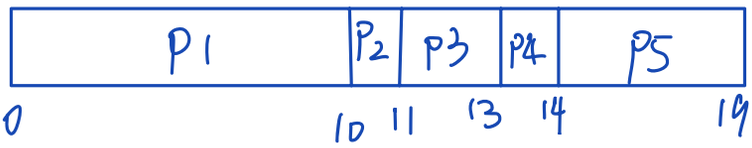
a. Draw four Gantt charts illustrating the execution of these processes using FCFS, SJF, a nonpreemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1) scheduling.

b. What is the turnaround time of each process for each of the scheduling algorithms in part a?

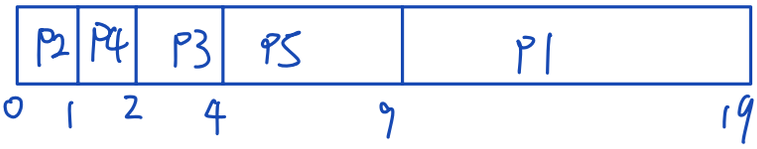
c. What is the waiting time of each process for each of the scheduling algorithms in part a?

d. Which of the schedules in part a results in the minimal average waiting time (over all processes)?

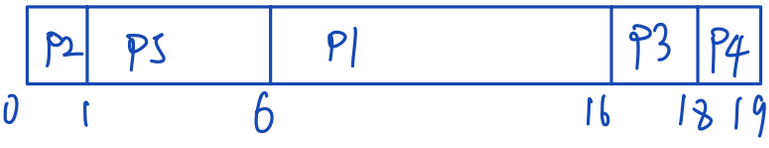
a. FCFS:



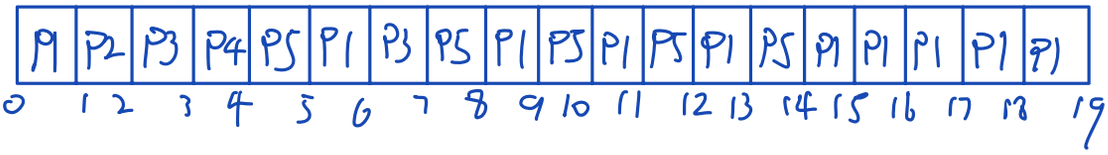
SJF:



priority:



RR:



b. FCFS: P1: 10ms P2: 11ms P3: 13ms P4: 14ms P5: 19ms

SJF: P1: 19ms P2: 1ms P3: 4ms P4: 2ms P5: 9ms

priority: P1: 16ms P2: 1ms P3: 18ms P4: 19ms P5: 6ms

RR: P1: 19ms P2: 2ms P3: 7ms P4: 4ms P5: 14ms

c: FCFS: P1: 0 P2: 10 P3: 11 P4: 13 P5: 14

SJF: P1: 9 P2: 0 P3: 2 P4: 1 P5: 4

priority: P1: 6 P2: 0 P3: 16 P4: 18 P5: 1

RR: P1: 9 P2: 1 P3: 5 P4: 3 P5: 9

d: Average waiting time:

FCFS: 9.6ms SJF: 3.2ms priority: 8.2ms RR: 5.4ms

So SJF results in the minimal average waiting time.